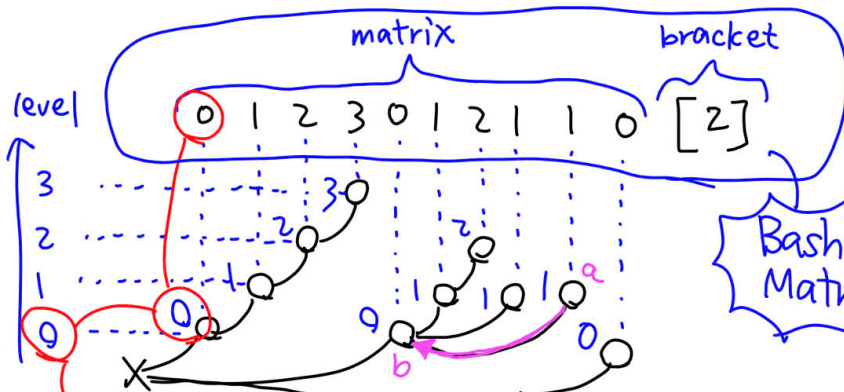


What is Bashicu Matrix?

- ① A bashicu Matrix consists of matrix and bracket.

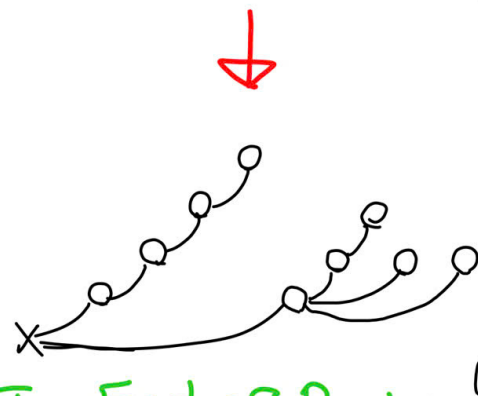
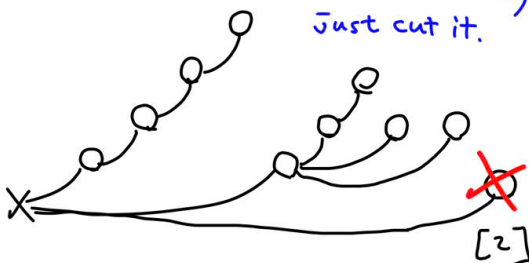


Notes

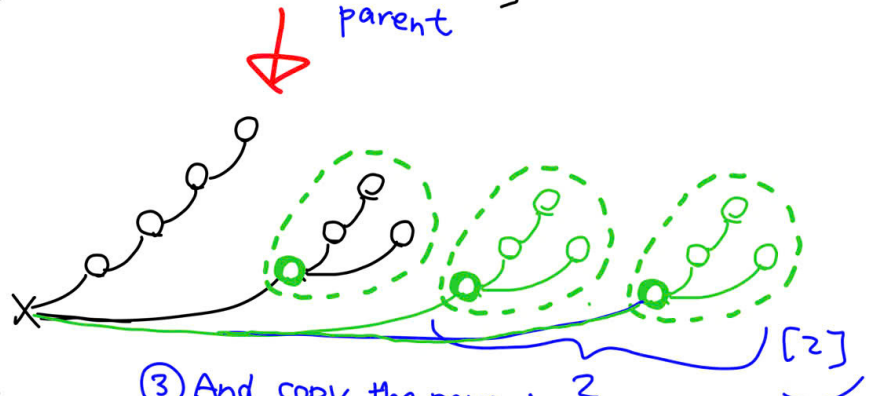
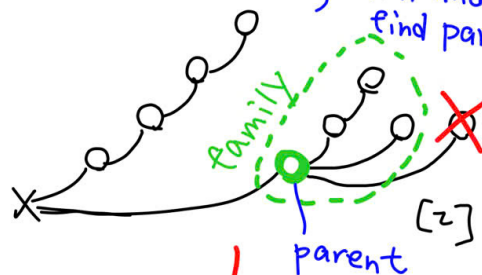
- ★ the true Bashicu Matrix has multiple rows. this is only 1 row Matrix.
- ★ 1 row Bashicu Matrix is called as Primitive Sequence.
- ★ This Bashicu Matrix and the rule for expansion is called as "Bashicu Matrix System".
- ★ The bracket of the true Bashicu Matrix System grows as $[n] \rightarrow [n^2]$ before each expansion.
- ★ Bashicu is the name of author of Bashicu Matrix System.
- ★ Bashicu Matrix System is called as BMS.
- ★ The author of this page is Koteitan.

Bashicu Matrix Expansion

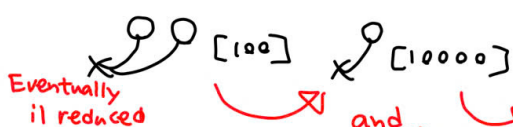
- ① If the rightmost node is zero, just cut it.



- ② If the rightmost node is NOT zero, cut it and find parent.



The End of Bashicu Matrix...



and... gone... this is the large Number. 1000000000

- ③ And copy the parent 2 and its family in the bracket times.

The Introduction of Multi-rows Bashicu Matrix System

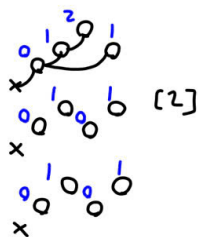
① The multi-rows BMS is like that.

$$\begin{pmatrix} 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} [2]$$

② It can be written in Multi-trees.

The top row is written in the same method of 1-row BMS.

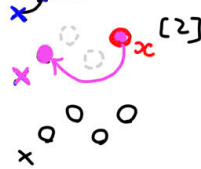
Write node for 2nd or lower row in the same method. The edges for the lower row is little bit difficult.



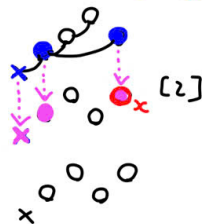
③ select a node.



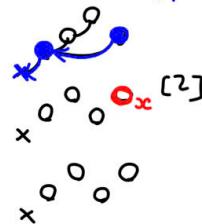
⑦ write an edge from x to $x-1$ with ignoring unmarked nodes.



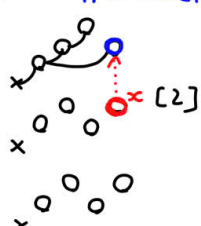
⑥ mark the nodes under the ancients.



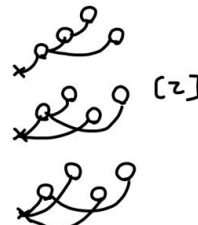
⑤ mark all ancients.



④ see upper node.

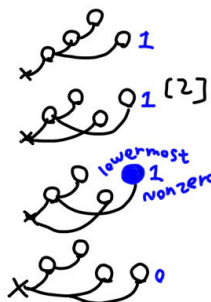


⑧ write all edges with this method.

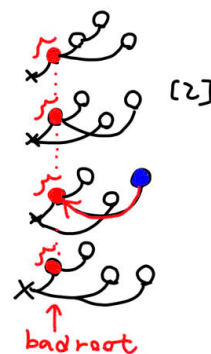


The Expansion of the Multi-rows Bashicu Matrix

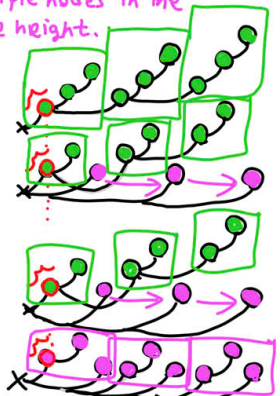
⑨ Find the lowermost nonzero in the rightmost column.



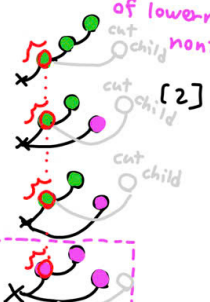
⑩ The column of the parent of lowermost nonzero is Bad root.



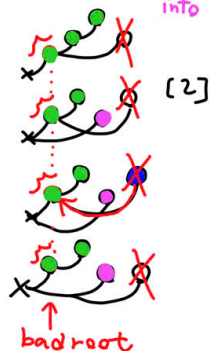
⑬ copy green nodes with ascension in the relative height difference between cut children and the bad root. copy purple nodes in the same height.



⑫ Mark all nodes in the rows under the row of lowermost nonzero.



⑪ mark all descendants of bad root into green. mark the left nodes into purple.



Notes

* The bracket grows $[n] \rightarrow [n^2]$ before the all expansions in the true Bashicu Matrix System.

* 1-row BMS is called as Primitive Sequence System, or PrSS.

* 2-rows BMS is called as Pair Sequence System or PSS.